Synthetic Maintenance and Safety Guide

Page 1: Unexpected Restart Review

After an unexpected controller restart, preserve evidence before making changes so the event can be traced.

- Record the restart time, controller status and any diagnostic messages that are still available.
- Review power-supply events, recent maintenance activity and configuration changes around the restart time.
- Confirm that the current application version matches the approved baseline before returning equipment to normal operation.
- Escalate repeated or unexplained restarts with captured diagnostics rather than repeatedly cycling power.

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Page 2: Firmware Change Management

Firmware changes should be treated as controlled engineering changes rather than routine troubleshooting shortcuts.

- Record the current firmware version, target version, device identifier and reason for the proposed change.
- Check compatibility with the engineering software, hardware revision and application requirements.
- Back up the approved configuration and define a rollback plan before the change.
- Document validation results after the change and update the maintenance record.

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Page 3: Safety Interlocks and Emergency Stop

Safety functions require independent review and must not be bypassed merely to make troubleshooting faster.

- Do not defeat or bypass an emergency-stop circuit or safety interlock as a diagnostic shortcut.
- Follow the site safety procedure and involve appropriately authorised personnel before testing safety-related functions.
- Where a safety function is implicated, preserve the observed state and relevant diagnostics before reset when safe to do so.
- Verify required safety functions according to the approved procedure before equipment is returned to service.

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